Cambridge Assessment International Education

Cambridge O Level

ADDITIONAL MATHEMATICS **4037/22**

Paper 2 **October/November 2023**

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mathematics-Specific Marking Principles

1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.

2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.

3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.

4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).

5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.

6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method marks, awarded for a valid method applied to the problem.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.

B Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more 'method' steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation 'dep' is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt answers which round to
cao correct answer only
dep dependent
FT follow through after error
isw ignore subsequent working
nfww not from wrong working
oe or equivalent
rot rounded or truncated
SC Special Case
soi seen or implied

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Question	Answer	Marks	Guidance
1(a)	$y = -\frac{1}{2}x + 25$ isw	**3**	**M1** for $m = \frac{29-23}{-8-4}$ oe or $-\frac{1}{2}$ and **M1 FT** for $\frac{y-23}{x-4} = their\left(-\frac{1}{2}\right)$ oe or $y = their\left(-\frac{1}{2}\right)x + c$ and $23 = -\frac{1}{2} \times 4 + c$ oe **OR** **M1** for solving $23 = 4m + c$ $29 = -8m + c$ for $m = -\frac{1}{2}$ or $c = 25$ and **M1 FT** for correctly using *their m* or *their c* to find *c* or *m*
	Solves *their* linear equation simultaneously with $y = 2x + 5$ to find x or y	**M1**	**FT** *their* $y = -\frac{1}{2}x + 25$ oe
	(8, 21)	**A1**	
1(b)	$\sqrt{8^2 + 21^2}$ oe	**M1**	**FT** *their* (8, 21)
	$\sqrt{505}$ isw or 22.5 or 22.47[22…] rot to 2 or more dp	**A1**	
2	$x^2 + 2kx = -2x - 6k - 1$	**M1**	
	$x^2 + (2k+2)x + 6k + 1[= 0]$	**A1**	
	Correctly uses $b^2 - 4ac$ [*0] for *their* equation $(2k + 2)^2 - 4(6k + 1)$ [*0]	**M1**	where * is any inequality sign or =; **FT** *their* 3-term quadratic in x and k
	$4k^2 - 16k$ [*0] nfww	**A1**	
	$k = 4$	**A1**	**dep** on all previous marks awarded

Question	Answer	Marks	Guidance
2	**Alternative method**		
	$2x + 2k$	**(M1)**	
	$k = -x - 1$ or $x = -k - 1$ oe	**(A1)**	
	$-2(-k-1) - 6k - 1 = (-k-1)^2 + 2k(-k-1)$ oe or $-2x - 6(-1-x) - 1 = x\,(x + 2(-1-x))$ oe	**(M1)**	**FT** *their k* of the form $ax + b$ where a and b are non-zero constants or *their x* of the form $ck + d$ where c and d are non-zero constants
	$k^2 - 4k$ [= 0] or $x^2 + 6x + 5$ [= 0] and $x = -5$ [$x = -1$] nfww	**(A1)**	
	$k = 4$	**(A1)**	**dep** on all previous marks awarded
3	$\dfrac{16+9\sqrt{3}}{(2+\sqrt{3})^2}$	**B1**	
	$\dfrac{(16+9\sqrt{3})(7-4\sqrt{3})}{(7+4\sqrt{3})(7-4\sqrt{3})}$ or $\dfrac{16+9\sqrt{3}}{7+4\sqrt{3}} \times \dfrac{7-4\sqrt{3}}{7-4\sqrt{3}}$	**M1**	**FT** $\dfrac{c(16+9\sqrt{3})}{a+b\sqrt{3}}$ where a, b and c are non-zero constants
	$112 - 64\sqrt{3} + 63\sqrt{3} - 108$ or $\dfrac{-112 + 64\sqrt{3} - 63\sqrt{3} + 108}{-1}$	**A1**	
	$4 - \sqrt{3}$ or $-\sqrt{3} + 4$ cao, nfww	**A1**	
	Alternative method		
	$\dfrac{16+9\sqrt{3}}{(2+\sqrt{3})^2}$	**(B1)**	
	$\dfrac{(16+9\sqrt{3})(2-\sqrt{3})^2}{(2+\sqrt{3})^2(2-\sqrt{3})^2}$ or $\dfrac{16+9\sqrt{3}}{(2+\sqrt{3})^2} \times \dfrac{(2-\sqrt{3})^2}{(2-\sqrt{3})^2}$	**(M1)**	
	$112 - 64\sqrt{3} + 63\sqrt{3} - 108$ or $64 - 32\sqrt{3} - 32\sqrt{3} + 48 + 36\sqrt{3} - 54 - 54 + 27\sqrt{3}$	**(A1)**	
	$4 - \sqrt{3}$ or $-\sqrt{3} + 4$ cao, nfww	**(A1)**	

Question	Answer	Marks	Guidance
4(a)	$\frac{e^{2x+2}}{e^{\frac{x}{2}}} = 10$ oe, soi	**B1**	
	$e^{1.5x+2} = 10$ oe	**M1**	**FT** $\frac{e^{2x+k}}{e^{\frac{x}{2}}} = 10$ oe or $\frac{e^{kx+2}}{e^{\frac{x}{2}}} = 10$ oe where k is an integer and $k > 0$ or $\frac{e^{2x+2}}{e^{\frac{x}{n}}} = 10$ oe where n is an integer and $n > 1$ or $n = -2$
	$1.5x + 2 = \ln 10$ oe	**M1**	**FT** an expression of, or equivalent to, the form $e^{ax+b} = 10$ oe where a and b are non-zero constants
	$x = \frac{2}{3}(\ln 10 - 2)$ oe, isw or 0.202 or 0.2017[23…] rot to 4 or more dp isw	**A1**	
4(b)	$\frac{y^2}{4y-9} = 9^{\frac{1}{2}}$ nfww or $\log_9 \frac{y^2}{4y-9} = \log_9 9^{\frac{1}{2}}$ oe	**M2**	**M1** for at least one correct log law used in a correct equation e.g. $\log_9 y^2 - \log_9(4y-9) = \frac{1}{2}$ or $\log_9 \frac{y^2}{4y-9} = \frac{1}{2}$ or $2\log_9 y - \log_9(4y-9) = \frac{1}{2}\log_9 9$
	$y^2 - 12y + 27$ [= 0] nfww	**A1**	
	$(y-3)(y-9) = 0$	**DM1**	**dep** on at least **M1** previously awarded
	$y = 3$, $y = 9$ nfww	**A1**	
5(a)	Correct first derivative: $3x^2 - 14x + 12$	**M2**	**M1** for two terms of $x^3 - 7x^2 + 12x - 5$ differentiated correctly
	[At $x = 1$] gradient of tangent: 1	**A1**	
	$y - 1 =$ *their*$(-1)(x - 1)$ oe or $y = -x + c$ **and** $1 = -1 + c$ soi	**M1**	**FT** $\frac{-1}{\textit{their}\left.\frac{dy}{dx}\right\|_{x=1}}$
	$y - 1 = -1(x - 1)$ or $y = -x + 2$ oe, isw	**A1**	

Question	Answer	Marks	Guidance
5(b)	$x^3 - 7x^2 + 12x - 5 = their(-x + 2)$	**B1**	**FT** *their* $y = ax + b$ where a is a non-zero constant
	Uses the correct linear factor $x - 1$ and the correct cubic $x^3 - 7x^2 + 13x - 7[=0]$ to find a quadratic factor with at least two terms correct	**M1**	
	$x^2 - 6x + 7$	**A1**	
	Correct use of formula or completing the square on *their* 3-term quadratic, e.g., $x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4[1](7)}}{2[1]}$ or $x = \frac{6 \pm \sqrt{36 - 4[1](7)}}{2}$	**M1**	**FT** *their* 3-term quadratic providing it is from an attempt at finding a quadratic factor and the discriminant is not negative
	$x = 3 \pm \sqrt{2}$	**A1**	
6	$\left[\frac{(x+2)^2}{x} =\right] \quad x + 4 + \frac{4}{x}$ soi	**B2**	**B1** for two terms correct or for $\frac{x^2 + 4x + 4}{x}$
	$\left[\frac{x^2}{2} + 4x + 4\ln x\right]_2^3$	**B2**	**B1** for $\left[\frac{x^2}{2} + ... + 4\ln x\right]_2^3$ or $[... + 4x + 4\ln x]_2^3$ or $\left[\frac{x^2}{2} + 4x + k\ln x\right]_2^3$ with $k \neq 0$
	$\left[\frac{9}{2} + 12 + 4\ln 3\right] - \left[\frac{4}{2} + 8 + 4\ln 2\right]$	**M1**	**dep** on at least previous **B1** for integration
	$6.5 + 4\ln\left(\frac{3}{2}\right)$ or exact equivalent	**A1**	
7(a)	Velocity: $3e^{2t} - 4e^{-2t} - 1$ isw	**B2**	**B1** for $3e^{2t}$ or $-4e^{-2t}$
	Acceleration: $6e^{2t} + 8e^{-2t}$ isw	**B1**	**FT** $me^{2t} + ne^{-2t} + k$ where m, n and k are constants

Question	Answer	Marks	Guidance
7(b)	$3e^{4t} - e^{2t} - 4 = 0$ or $3\left(e^{2t}\right)^2 - e^{2t} - 4 = 0$	**B1**	
	$\left(3e^{2t} - 4\right)\left(e^{2t} + 1\right) = 0$	**M1**	**FT** *their* 3-term quadratic in e^{2t} oe
	$e^{2t} = \frac{4}{3}$ nfww	**A1**	
	$\frac{1}{2}\ln\frac{4}{3}$ oe, isw or 0.144 or 0.1438[41…] rot to 4 or more dp and no other solutions	**A1**	
7(c)	$6 \times e^{2\left(\frac{1}{2}\ln\frac{4}{3}\right)} + 8 \times e^{-2\left(\frac{1}{2}\ln\frac{4}{3}\right)}$	**M1**	**FT** $pe^{2t} + qe^{-2t}$ where p and q are non-zero constants and *their* **positive** $\frac{1}{2}\ln\frac{4}{3}$ from part (b)
	14 nfww	**A1**	
8(a)	Derivative of $\sin 2x$: $2\cos 2x$ soi	**B1**	
	Product rule: $x \times 2\cos 2x + [1]\sin 2x$ isw	**B1**	**FT** *their* $2\cos 2x$
8(b)	$y = \frac{\pi}{4}$ soi, isw	**B1**	
	gradient of tangent: 1 soi	**B1**	**dep** on correct derivative
	$y = x$ or $y - x = 0$ or $x - y = 0$	**B1**	**dep** on correct derivative
8(c)	$\left[x\sin 2x + \frac{1}{2}\cos 2x\right]_0^{\frac{\pi}{6}}$ nfww	**M3**	**M2** for $x\sin 2x + k\cos 2x$ where $k > 0$ or $k = -\frac{1}{2}$; nfww or **M1** for $\int 2x\cos 2x\,dx = x\sin 2x - \int \sin 2x\,dx$ or $\frac{-\cos 2x}{2} + \int 2x\cos 2x\,dx = x\sin 2x$
	$\frac{\pi}{6}\sin\frac{\pi}{3} + \frac{1}{2}\cos\frac{\pi}{3} - \frac{1}{2}\cos 0$	**A1**	
	$\frac{\pi\sqrt{3}}{12} - \frac{1}{4}$ or $\frac{\pi\sqrt{3}-3}{12}$	**A1**	

Question	Answer	Marks	Guidance
9(a)	Correct pair of simplified linear equations in a and d with terms collected, e.g., $3a+3d=-36$ isw or $a+d=-12$ isw $3a+30d=72$ isw or $a+10d=24$ isw	**B3**	**B2** for one correct simplified equation or **B1** for $a+a+d+a+2d=-36$ or $\frac{3}{2}\{2a+(3-1)d\}=-36$ or $a+9d+a+10d+a+11d=72$ or $\frac{12}{2}\{2a+(12-1)d\}-\frac{9}{2}\{2a+(9-1)d\}=72$ or $12a+66d-9a-36d=72$ or $\frac{3}{2}\{2(a+9d)+(3-1)d\}=72$
	Solves two linear equations for d or a e.g. $27d=108 \rightarrow d=\ldots$ or $9d=36 \rightarrow d=\ldots$ or $a+10(-12-a)=24 \rightarrow a=\ldots$ $27a=-432 \rightarrow a=\ldots$	**M1**	**FT** *their* linear equations in a and d providing at least **B1** earned and the equations have a solution
	$d=4$ and $a=-16$ nfww	**A1**	
9(b)	1.2^n *101	**B3**	where * is any inequality sign or =; **B2** for $\frac{[1](1.2^n-1)}{(1.2-1)}*500$ or **B1** for $r=1.2$ soi
	$n\log 1.2*\log 101$ or $\log_{1.2}101$ soi	**M1**	**FT** 1.2^n * *their* 101 providing **B2** has been awarded and (*their* 101) > 0
	$n=26$	**A1**	**dep** on all previous marks awarded

Question	Answer	Marks	Guidance
10(a)	Writes cot$x$ and tan$x$ in terms of sin$x$ and cos$x$: $\frac{\sin x}{1-\frac{\cos x}{\sin x}}+\frac{\cos x}{1-\frac{\sin x}{\cos x}}$	**M1**	OR $\frac{\sin x\left(1-\frac{\sin x}{\cos x}\right)+\cos x\left(1-\frac{\cos x}{\sin x}\right)}{\left(1-\frac{\cos x}{\sin x}\right)\left(1-\frac{\sin x}{\cos x}\right)}$
	Simplifies denominator: $\frac{\sin x}{\frac{\sin x-\cos x}{\sin x}}+\frac{\cos x}{\frac{\cos x-\sin x}{\cos x}}$	**A1**	OR $\frac{\sin x\left(\frac{\cos x-\sin x}{\cos x}\right)+\cos x\left(\frac{\sin x-\cos x}{\sin x}\right)}{\left(\frac{\sin x-\cos x}{\sin x}\right)\left(\frac{\cos x-\sin x}{\cos x}\right)}$
	Writes as two simple algebraic fractions: $\frac{\sin^2 x}{\sin x-\cos x}+\frac{\cos^2 x}{\cos x-\sin x}$	**A1**	OR writes as a single simple algebraic fraction: $\frac{\sin^2 x(\cos x-\sin x)+\cos^2 x(\sin x-\cos x)}{(\sin x-\cos x)(\cos x-\sin x)}$
	Writes as a difference with a common denominator: $\frac{\sin^2 x}{\sin x-\cos x}-\frac{\cos^2 x}{\sin x-\cos x}$	**A1**	OR $\frac{\sin^2 x(\cos x-\sin x)-\cos^2 x(\cos x-\sin x)}{(\sin x-\cos x)(\cos x-\sin x)}$
	Correct simplification to given answer, e.g., $\frac{(\sin x-\cos x)(\sin x+\cos x)}{(\sin x-\cos x)}=\sin x+\cos x$ or $\frac{\cancel{(\sin x-\cos x)}(\sin x+\cos x)}{\cancel{(\sin x-\cos x)}}\left[=\sin x+\cos x\right]$	**A1**	All steps correct and final step fully justified by factorising

Question	Answer	Marks	Guidance
10(b)	$10\cos^2 x + 3\cos x - 1 [= 0]$ or $\sec^2 x - 3\sec x - 10 [= 0]$	**B2**	**B1** for $\frac{9\cos x}{\sin x} + \frac{3}{\sin x} = \frac{\sin x}{\cos x}$ or better or $9 + \frac{3\tan x}{\sin x} = \tan^2 x$ or better OR **M1** for one sign error in $10\cos^2 x + 3\cos x - 1 [= 0]$ or $\sec^2 x - 3\sec x - 10 [= 0]$
	$(5\cos x - 1)(2\cos x + 1)[= 0]$ or $(\sec x - 5)(\sec x + 2)[= 0]$	**M1**	**FT** *their* 3-term quadratic in cos*x* or sec*x*
	[cos*x* = $\frac{1}{5}$ **and** cos*x* = $-\frac{1}{2}$ OR sec*x* = 5 **and** sec*x* = –2 leading to] 78.5 or 78.46[30…] rot to 2 or more dp 281.5 or 281.53[69…] rot to 2 or more dp 120 240 and no extras in range $0 < x < 360$	**A2**	**A1** for any two correct angles [found using cos*x* = $\frac{1}{5}$ **and** cos*x* = $-\frac{1}{2}$ OR sec*x* = 5 **and** sec*x* = –2]; ignore extras